# Pagemark - Google Play audit and publishing guide

# Google Play readiness - Pagemark 0.7.0

Static review of manifest, Gradle, dependencies and docs (2026-10-12). Play policy details change; confirm each item in Play Console before relying on it. Nothing was uploaded or registered.

## Verdict: the code blockers are fixed (2026-10-12); what is left needs your Google account.

Update after 0.7.1: the `play` flavor and `tool/build.ps1` exist and produce a signed `.aab`; the merged manifest has no installer permission; the HIGH findings are fixed in code and covered by tests. Remaining items are listed in ../PLAY_STORE.md (account, closed test, OAuth SHA-1, feature graphic, Data safety form, real-device check).

Original assessment below.

| # | Item | Status | Action |
|---|---|---|---|
| 1 | **Self-updater** (`REQUEST_INSTALL_PACKAGES`, `open_filex`, `updater.dart`) | BLOCKER | Play does not allow an app to download and install APKs outside Play. Build a Play flavor with the permission and Updates screen removed; keep the GitHub-APK build for sideloaders. |
| 2 | **App Bundle (.aab)** | BLOCKER (process) | Play takes AAB, not APK. `flutter build appbundle` is not set up or tested; no release script exists (PGM-222). |
| 3 | Signing | Needs decision | Play App Signing re-signs with Google's key. Your keystore becomes the *upload* key. Sideload APKs (your key) and Play installs (Google key) cannot update each other. Keep the keystore backed up. |
| 4 | Google Drive sign-in | Needs work | The OAuth Android client is tied to a SHA-1. Add the Play App Signing SHA-1 to the Cloud project, or Drive backup breaks on Play installs. `drive.appdata` is a non-sensitive scope. |
| 5 | Developer account | Needs you | Play Console requires a one-time registration fee (about $25) and identity verification. Personal accounts created recently must run a closed test (about 12 testers for 14 days) before production. This is a Google requirement, not something I am suggesting you buy beyond it. |
| 6 | targetSdk | OK | 36, meets the current requirement. Recheck each August. |
| 7 | 16 KB page size | OK | Native libs verified aligned (PGM-226). |
| 8 | Privacy policy URL | Mostly done | `docs/PRIVACY.md` exists in the public repo; Play needs a public URL (the GitHub page works) and it must match the Data safety form. Fix PGM-201/203/243 first so the policy is true. |
| 9 | Data safety form | To do | Declare: no data collected by you; Drive backup goes to the user's own Drive; Open Library lookups send ISBN/title text; no analytics. |

| # | Item | Status | Action |
|---|---|---|---|
| 10 | Permissions | OK after #1 | CAMERA (scanner), USE_BIOMETRIC, POST_NOTIFICATIONS, INTERNET are all justifiable. Remove REQUEST_INSTALL_PACKAGES. |
| 11 | Store listing | To do | 512 px icon, 1024x500 feature graphic, 2+ phone screenshots (you have real ones in docs/screenshots), short and full description, content rating questionnaire (expect Everyone; the optional heat/content-notes field may raise a question, answer honestly). |
| 12 | Crashes/quality | Watch | Pre-launch report from Play will run robo tests; the lock screen and FLAG_SECURE may block it, so provide test steps or a test PIN. |
| 13 | Open HIGH findings | Fix first | PGM-100, 101, 102, 200, 202 affect trust claims a store listing would repeat. |
| 14 | Doc truth | Fix first | PGM-240..243, 248 (stale/false claims). |
| 15 | Versioning | OK | versionCode is monotonic; Play needs a unique, increasing code per upload. |

## Suggested order

1. Fix PGM-100/101/102/200/202 and the false doc claims (a 0.7.1 or 0.8.0).
2. Add a `play` build flavor: no installer permission, no Updates screen. Build a signed AAB and test it on the emulator and a phone.
3. Create the Play Console account, upload the AAB to internal testing, enrol Play App Signing, copy its SHA-1 into the Drive OAuth client.
4. Run a closed test, fill the Data safety form and store listing, then apply for production.

# Publishing Pagemark on Google Play

Everything the repo can prepare is here; the steps that need your Google account are marked **You**. Verify each answer in Play Console before submitting, because Google changes its forms.

## Build

```
powershell -File tool\build.ps1
```

Produces `build/release-<version>/Pagemark-<version>-play.aab` (upload to Play) and the two GitHub APKs. The bundle is signed with your release keystore, which becomes the **upload key**. Google re-signs what users download with its own **app signing key**, so:

- GitHub-APK installs (your key) and Play installs (Google's key) cannot update each other. Pick one per phone.
- Keep `~/.android/pagemark-release.jks` backed up. Losing it means asking Google to reset the upload key.

The `play` flavor has no `REQUEST_INSTALL_PACKAGES` permission and no Updates screen (checked in the merged manifest).

## Steps

1. **You:** create a Play Console developer account (one-time registration fee, identity verification). A personal account created recently must run a **closed test with about 12 testers for 14 days** before production access.
2. **You:** create the app (name Pagemark, free, no ads), opt in to **Play App Signing**, upload the `.aab` to **Internal testing** first.
3. **You:** Play Console > Setup > App signing shows the **SHA-1 of Google's app signing key**. Add an Android OAuth client with that SHA-1 (package `com.axialforge.pagemark`) in your Google Cloud project, next to the debug and release ones in DRIVE_SETUP.md. Without it, Drive sign-in fails in the Play build.
4. Fill in the forms below, run the closed test, then apply for production.

## Store listing

- **Short description (80):** Your private reading log: print, e-book and audiobook, all on your phone.
- **Full description:** Pagemark is a book tracker that keeps everything on your phone. Log print, e-book and audiobook reads, rate half stars, write private thoughts, save quotes, set a yearly goal and see your stats. Scan a barcode or search to add books (covers and details come from Open Library). Keep a locked library behind a PIN or your fingerprint. Back up to your own Google Drive on a schedule, optionally encrypted with a passphrase you choose. Export CSV/JSON, import from Goodreads or StoryGraph. No account, no ads, no tracking.
- **Category:** Books & Reference. **Tags:** reading tracker, book log.
- **Graphics:** 512x512 icon (`assets/branding`), 1024x500 feature graphic (**You**: not yet made), at least 2 phone screenshots (use `docs/screenshots/`, regenerate for 0.7.1 first: several images are version-stamped 0.4.1-0.5.0).
- **Privacy policy URL:** `https://github.com/AxialForge/Pagemark/blob/main/docs/PRIVACY.md` (public; keep it matching the app).

## Data safety form (answers)

- Does the app collect or share user data? **Pagemark itself collects nothing and has no server.** Data you choose to send goes to services you control or public lookups:
- **Google Drive backup (optional):** your library file is uploaded to the app folder of your own Drive. Declare as data shared with Google Drive at the user's request; encryption in transit: yes; user can request deletion: yes (delete from Drive / sign out).
- **Book lookup:** the title, author or ISBN you type or scan is sent to Open Library. Declare as "search query / app activity", not linked to identity, optional.
- Camera: used only to read barcodes on the device; images are not stored or sent.
- No ads, no analytics, no crash-reporting SDK. Data is encrypted in transit (HTTPS). Users can delete data by clearing app storage or deleting backups.

## Content rating

Expect **Everyone**. The optional "heat level and content notes" field is a private diary field, off by default, with no content provided by the app. Answer the questionnaire honestly (user-generated content: none shared with other users).

## Other declarations

- Target audience: 13+ (not designed for children).
- Permissions: Camera (scanner), Biometric (lock), Notifications (backup problems, Android 13+), Internet. Foreground-service permissions come from the background-backup library for short tasks; if Play asks for a foreground service declaration, describe "short scheduled upload of the user's backup".
- Pre-launch report: Play's test robot cannot get past the lock screen if one is set; leave the app lock off in the test build.

## Before you submit (open items)

- Verify the 0.7.1 fixes on a real phone (see `docs/release-audit/MANUAL_TEST_PLAN.md`).
- Regenerate screenshots and the PDF manuals for the release you ship.
- Make the 1024x500 feature graphic.